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Statistical tests for VaR and CVaR

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Title: Statistical tests for VaR and CVaR

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Abstract: The thesis presents test statistics of Value-at-Risk and Conditional Value-at-Risk. The reader is familiar with basic nonparametric estimators and their asymptotic distributions. Tests of accuracy of Value-at-Risk are explained and asymptotic test of Conditional Value-at-Risk is derived. The thesis is concluded by process of backtesting of Value-at-Risk model using real data and computing statistical power and probability of Type I error for selected tests.

Keywords: Value-at-Risk, VaR, Conditional Value-at-Risk, CVaR, Test statistic, Backtesting

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Introduction

The thesis deals with statistical tests of Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR). They are the most often used quantitative risk measures by banks and insurance companies as economical indicators of potential loss. The thesis is divided into 5 chapters.

Chapter 1 describes the history and main advantages and disadvantages of VaR and CVaR.

In Chapter 2, the most known nonparametric estimators of VaR and CVaR are defined and their asymptotic distributions are presented. In this chapter, other methods of calculating VaR are discussed.

In Chapter 3, tests of accuracy of VaR are presented. Unconditional coverage tests, independence tests and join test are explained and some of their shortcomings are stated. The traffic light coverage test is presented as the test given by the regulatory framework. Finally, asymptotic test for location of CVaR is derived.

Chapter 4 shows the process of backtesting the model of Value at Risk on three different financial instruments. All previously presented tests of VaR are used and results are afterwards discussed.

In Chapter 5, the statistical power and probability of Type I error is computed for three selected tests and three different distributions. Results are showed in plots and are discussed.

1. About VaR and CVaR

Value-at-Risk¹ is the most popular quantitative risk measure used by banks and insurance companies. This economical indicator provides an estimate of the potential loss of a given portfolio. Therefore, it is a statistical estimate which provides the highest loss for the probability α and over a given time horizon.

A disadvantage of VaR is that we have no information about possible maximal losses in case of exceeding this value. That's why some use the Conditional Value-at-risk². It provides the average of potential loss that exceeds the VaR.

1.1 History

The first reference to VaR is dated back to the 1940s when Leavens (1945) published an example of a portfolio containing ten bonds. He assumed that each bond would mature for \$ 1,000 or default and be worthless. Individual defaults were assumed as independent to each other. In this example, he didn't mention VaR but frequently spoke about "spread between probable losses and gains". Probably, he had the standard deviation of the market portfolio in mind. At the end, this portfolio had VaR \$ 948.69.

The next mention about VaR was published independently by both Markowitz (1952) and Roy (1952). They tried to develop a way how to select a portfolio that optimizes reward for a given level of risk. In spite of the fact that the covariances between risk factors and diversification effects were incorporated into their VaR measures and both seemed mathematically similar, both returned very different results. Roy's VaR measure required covariances which had to be estimated from past information. On the second hand, Markowitz's VaR measure requires only a covariance matrix for risk factors that isn't necessary to be estimated from empirical data. Furthermore, he argued that statistical methods should be combined with common sense.

From the 1950s to the 1970s, many VaR models were published. Because of the low processing power, VaR measures were more likely theoretical. Some articles were written by Tobin (1958), by Lintner (1965) and by Mossin (1966). The large majority of VaR models were aimed at Portfolio theory. In the case when these models were used for assets, the number of modelling issues increased and models became very inefficient. Schrock (1971) and Dusak (1973) described simple VaR measure for futures portfolio, but they didn't take into account the seasonality. However, because primarily agricultural products were traded, the seasonality was an integral part of it. So these VaR measures were useless. Lietaer (1971) described VaR measure for foreign exchange rate. Since World War II most currencies had devalued. Some of them had done several times. Governments tried to cover plan to devalue, so Lietaer came with a model on how corporations could optimize such hedges. He modified a model which was proposed by Sharpe (1963). In Lietaer's book, Monte Carlo simulations were firstly used.

¹abbreviated as VaR or V@R

²abbreviated as CVaR or CV@R and equivalent to the Expected Shortfall (ES), Tail Value-at-Risk (TVaR), Tail Conditional Expectation (TCE), Average Value-at-Risk (AVaR) and Expected Tail Loss (ETL)

In the 1970s and the 1980s, technology improved, so models published before started to be more useful. Also new financial products and new risks appeared and it was necessary to model them and prevent them. For example in 1971, when the Bretton Woods agreement collapsed, floating exchange rates were allowed and the foreign exchange forward market started to grow rapidly. Today it is the largest forward market in the world.

On June 26, 1974, the German government had to force the bankruptcy of Bank Herstatt. That day many of other German banks provided payments from DEM³ to USD⁴ which had to be delivered to New York in the USA. Because of different time-zones, American banks were closed and Herstatt's bankruptcy occurred before they opened, all German banks didn't receive their USD payments and lost lots of money. This event led G-10⁵ countries to form a committee under the auspices of the Bank for International Settlements (BIS⁶). This committee was called the Basel Committee on Banking Supervision and its members were representatives from central banks and regulatory authorities. In 1988, the Basel Committee published a set of minimal capital requirements for banks, called Basel I, which required banks to report and quantify credit risk. In 2004, a second set of requirements, called Basel II, were published. The Basel committee demanded banks to report and quantify market risk. Last published set of requirements from 2009, called Basel III, requires report and quantify operational risk. Nowadays banks must cover three times the 99% VaR of possible lost over 10 days by equity.

1.2 Advantages and disadvantages

Artzner et al. (1999) defined the coherence⁷ as a set of desirable properties of risk measure. They also showed that VaR is not coherent compared with CVaR, which is. VaR does not satisfies the assumption of sub-additivity. This means that the risk of portfolio could be higher than the sum of individual risks of this portfolio. Rockafellar and Uryasev (2000) showed that CVaR is convex, but VaR is not. The convexity is a very important property for solving optimization problems by linear and convex programming, so VaR is not well suited for it. The main disadvantage of CVaR over VaR is that it may not exist for heavy tailed distributions. The existence for CVaR is assumed only for distributions with finite second moment.

³code for the Deutsche Mark, former currency of Germany

⁴code for the United States dollar, official currency of the United States

⁵group of 11 countries: Belgium, Canada, France, Germany, Italy, Japan, Netherlands, Sweden, Switzerland, United Kingdom and United States

⁶BIS is an international financial institution owned by central banks which fosters international monetary and financial cooperation and serves as bank for central banks

⁷A risk measure satisfying the four axioms of translation invariance, sub-additivity, positive homogeneity and monotonicity is called coherent.

2. General picture of VaR and CVaR

In this chapter, I will assume that X is a random variable that represents a loss. All estimations will be based on historical data. Other methods showing how to calculate the Value-at-Risk will be presented later.

Definition 1. Let X be a continuous random variable with cumulative distribution function $F(x) = P[X \leq x], x \in \mathbb{R}$. Let α be a confidence level satisfying $\alpha \in (0, 1)$. Then VaR and CVaR are defined by formulas

$$\begin{aligned} \text{VaR}_\alpha(X) &= F^{-1}(\alpha) = \inf\{x \in \mathbb{R} : \alpha \leq F(x)\}, \\ \text{CVaR}_\alpha(X) &= E[X | X > F^{-1}(\alpha)]. \end{aligned}$$

In Figure 2.1 you can see example of VaR and CVaR, with $\alpha = 0.95$, of random variable X with lognormal distribution with $\mu=1$ and $\sigma=0.5$. The $\text{VaR}_\alpha(X)$ is 6.187 and the $\text{CVaR}_\alpha(X)$ is 14.166.

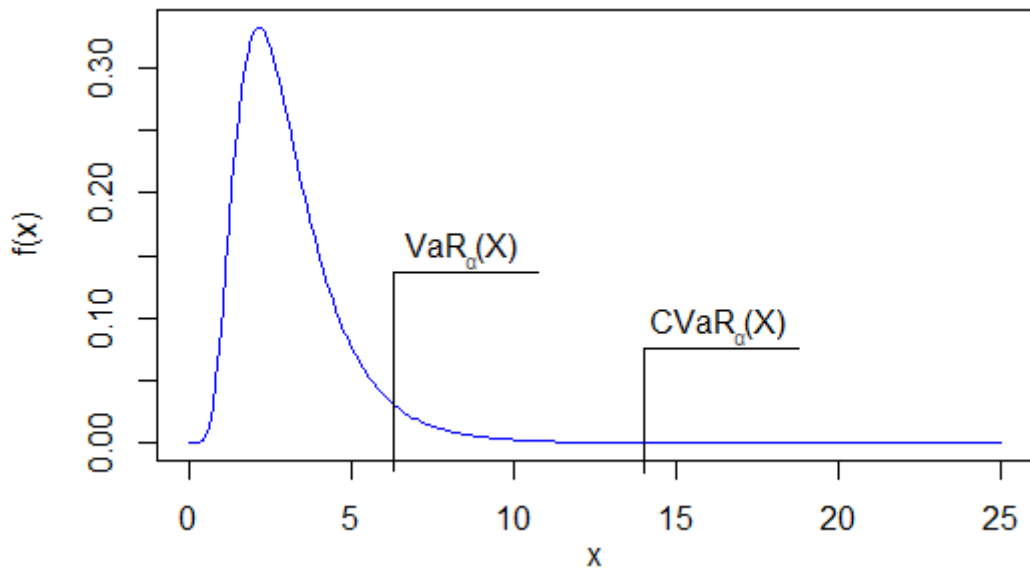


Figure 2.1: VaR and CVaR of lognormal distribution with $\mu=1$ and $\sigma=0.5$.

Now, after defining VaR and CVaR, I will show their estimators. In Definition 1 we can see that VaR is the α -quantile of loss given by a random variable X so I will look for an estimator of the α -quantile.

Let $X_1, X_2, \dots, X_n$ be i.i.d.¹ sample from continuous probability distribution with cdf² $F(x)$ and with pdf³ $f(x)$. Let $X_{(1)}, X_{(2)}, \dots, X_{(n)}$ be the same sample that is sorted in an ascending order, so that $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$. Then the most commonly used estimator for VaR and the α -quantile is the sampled quantile

$$\widehat{\text{VaR}}_\alpha(X) = \widehat{F}^{-1}(\alpha) = X_{(\lceil n\alpha \rceil)} \quad (2.1)$$

¹independent and identically distributed

²cumulative distribution function

³probability density function

and for CVaR is the tail sample mean

$$\widehat{\text{CVaR}}_\alpha(X) = \frac{1}{\lceil n(1-\alpha) \rceil} \sum_{i=\lceil n\alpha \rceil}^n X_{(i)}. \quad (2.2)$$

For simplicity, I will assume that α is chosen so that $n\alpha$ is an integer. Then the estimators for VaR and CVaR are

$$\begin{aligned} \widehat{\text{VaR}}_\alpha(X) &= X_{(n\alpha)}, \\ \widehat{\text{CVaR}}_\alpha(X) &= \frac{1}{n(1-\alpha)} \sum_{i=n\alpha}^n X_{(i)}. \end{aligned}$$

This assumption will simplify my following computations and will not have a negative impact because I will derive asymptotic results in large samples where the effect of rounding is negligible.

2.1 Distribution of estimator for VaR

The following theorem states the cdf and pdf of the ordered sample statistic $X_{(r)}$.

Theorem 1.

Let $X_1, X_2, \dots, X_n$ be i.i.d. sample from continuous probability distribution with cdf $F(x)$ and pdf $f(x)$ and $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ be corresponding ascending ordered sample. Let n be greater than 2 and r be integer, so that $1 \leq r \leq n$. Then cumulative distribution function of $X_{(r)}$ is

$$G_r(x) = \sum_{i=r}^n \binom{n}{i} F^i(x) (1 - F(x))^{n-i} \quad (2.3)$$

and probability density function of $X_{(r)}$ is

$$g_r(x) = n \binom{n-1}{r-1} f(x) F^{r-1}(x) (1 - F(x))^{n-r}. \quad (2.4)$$

Proof. See (Anděl; 2007, theorems 5.3 and 5.4). □

When n is large then Formulas (2.3) and (2.4) quickly become computationally difficult. Hence it is useful to use asymptotic distribution of VaR. Before deriving it, I state the Delta method for univariate random variable.

Theorem 2 (Delta method).

Let X_n be a sequence of random variables which satisfies for $\theta \in \mathbb{R}$ and $\sigma^2 \in \mathbb{R}$ that⁴

$$\sqrt{n}(X_n - \theta) \xrightarrow{D} N(0, \sigma^2), \quad n \rightarrow \infty.$$

Then

$$\sqrt{n}(g(X_n) - g(\theta)) \xrightarrow{D} N(0, \sigma^2[g'(\theta)]^2), \quad n \rightarrow \infty$$

for the function $g : \mathbb{R} \rightarrow \mathbb{R}$ which satisfies $g'(\theta) \neq 0$.

⁴ $\xrightarrow{D}$ denotes convergence in distribution

Proof. See (Davison; 2003, pages 33-35). □

Now, suppose that $X_1, \dots, X_n$ are i.i.d. continuous variables from distribution with cdf $F(x)$ and pdf $f(x)$. An equivalent way of defining $\widehat{\text{VaR}}_\alpha(X)$ in (2.1) is

$$\widehat{\text{VaR}}_\alpha(X) = F_n^{-1}(\alpha) = \inf\{x, F_n(x) \geq \alpha\}, \quad (2.5)$$

where $F_n(x)$ is empirical distribution function defined as

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[X_i \leq x], \quad x \in \mathbb{R}.$$

For fixed $y \in \mathbb{R}$, $F_n(y)$ is a random variable such that

$$\begin{aligned} \mathbb{E}[F_n(y)] &= F(y), \\ \text{var}[F_n(y)] &= F(y)(1 - F(y)) \end{aligned}$$

and using Central limit Theorem for fixed $y \in \mathbb{R}$

$$\sqrt{n}(F_n(y) - F(y)) \xrightarrow{D} N(0, F(y)(1 - F(y))), \quad n \rightarrow \infty.$$

As function $g(t)$ in Theorem 2 we define

$$g(t) = F^{-1}(t), \quad t \in (0, 1).$$

Using

$$\begin{aligned} z = F^{-1}(t) &\iff F(z) = t \implies f(z)dz = dt \implies \\ &\implies \frac{dy}{dt} = \frac{1}{f(y)} = \frac{1}{f(F^{-1}(t))} \end{aligned}$$

we have the first derivative of $g(t)$

$$g'(t) = \frac{d}{dt}(F^{-1}(t)) = \frac{1}{f(F^{-1}(t))}.$$

Therefore, using Delta method

$$\sqrt{n}(F^{-1}(F_n(y)) - F^{-1}(F(y))) \xrightarrow{D} N\left(0, \frac{F(y)(1 - F(y))}{[f(F^{-1}(F(y)))]^2}\right), \quad n \rightarrow \infty.$$

If we choose y as $\text{VaR}_\alpha(X)$ and realize that $F^{-1}(F_n(\text{VaR}_\alpha(X)))$ can be written as order statistic $X_{(n\alpha)} = \widehat{\text{VaR}}_\alpha(X)$, then

$$\sqrt{n}(\widehat{\text{VaR}}_\alpha(X) - \text{VaR}_\alpha(X)) \xrightarrow{D} N\left(0, \frac{\alpha(1 - \alpha)}{[f(\text{VaR}_\alpha(X))]^2}\right), \quad n \rightarrow \infty.$$

When we consider two sample VaRs of different confidence levels α and β , such that $0 < \alpha < \beta < 1$, it was shown by Ikeda and Matsunawa (1972) that they have normal asymptotic joint distribution with limiting correlation coefficient

$$\lim_{n \rightarrow \infty} \text{Corr}[\widehat{\text{VaR}}_\alpha(X), \widehat{\text{VaR}}_\beta(X)] = \sqrt{\frac{\alpha(1 - \beta)}{\beta(1 - \alpha)}}.$$

2.2 Distribution of estimator for CVaR

Manistre and Hancock (2005) showed that CVaR has normal asymptotic distribution and that for $\alpha \in (0, 1)$ holds

$$\sqrt{n}(\widehat{\text{CVaR}}_\alpha(X) - \text{CVaR}_\alpha(X)) \xrightarrow{D} N\left(0, \frac{\text{var}[X|X > \text{VaR}_\alpha(X)] + \alpha(\text{CVaR}_\alpha(X) - \text{VaR}_\alpha(X))^2}{1 - \alpha}\right), \quad n \rightarrow \infty.$$

They also provided the bivariate distribution for VaR and CVaR with confidence level $0 < \alpha < 1$. It follows bivariate normal asymptotic distribution with limiting covariance

$$\lim_{n \rightarrow \infty} n \text{Cov} \left[\widehat{\text{VaR}}_\alpha(X), \widehat{\text{CVaR}}_\alpha(X) \right] = \alpha \frac{\text{CVaR}_\alpha(X) - \text{VaR}_\alpha(X)}{f(\text{VaR}_\alpha(X))}.$$

2.3 Other nonparametric estimators

The estimation of VaR in Formula (2.5) is very simple and it is not computationally difficult. The disadvantage of this estimator is that it cannot extrapolate beyond the maximum observed data point. This extrapolation is needed when the sample is not too large and we suppose that the loss can be larger than the maximum from observed data. In this case, Alemany et al. (2013) recommends to use classical kernel estimation for cdf $F(x)$ instead of the empirical distribution function in (2.5). Then the estimator for $F(x)$ is

$$\begin{aligned} \widehat{F}(x) &= \int_{-\infty}^x \widehat{f}(u) du = \int_{-\infty}^x \frac{1}{nb} \sum_{i=1}^n k\left(\frac{u - X_i}{b}\right) du = \frac{1}{n} \sum_{i=1}^n \int_{-\infty}^{\frac{x - X_i}{b}} k(t) dt = \\ &= \frac{1}{n} \sum_{i=1}^n K\left(\frac{x - X_i}{b}\right), \end{aligned}$$

where $k(x)$ is a kernel function, b is a bandwidth, $X_1, \dots, X_n$ is random sample and $K(x)$ is a cdf of $k(x)$. Silverman (1986) presents the most common kernel functions which are the Epanechnikov and the Gaussian kernel. The estimation of $\text{VaR}_\alpha(X)$ from $\widehat{F}(x)$ we obtain, using the Newton-Raphson method, by solving an equation

$$\widehat{F}(x) = \alpha$$

and we denote it $\widehat{\text{VaR}}_\alpha^K(X)$.

Formula (2.2) of $\widehat{\text{CVaR}}_\alpha(X)$ can be rewritten equivalently as

$$\widehat{\text{CVaR}}_\alpha(X) = \frac{1}{\lceil n\alpha \rceil} \sum_{i=1}^n X_i \mathbf{1}[X_i \geq \widehat{\text{VaR}}_\alpha(X)]. \quad (2.6)$$

Scaillet (2004) proposed estimation of $\text{CVaR}_\alpha(X)$ by replacing $\widehat{\text{VaR}}_\alpha(X)$ and the indicator function in (2.6) as

$$\widehat{\text{CVaR}}_\alpha^K(X) = \frac{1}{n\alpha} \sum_{i=1}^n X_i K\left(\frac{\widehat{\text{VaR}}_\alpha^K(X) - X_i}{b}\right),$$

where $K(x)$ is cdf of kernel function $k(x)$ and b is a bandwidth.

The kernel smoothed version increases the precision of VaR estimates, but Chen (2008) argues that it does not increase the precision of CVaR estimate and recommends to use the tail sample mean. Another estimate of CVaR was proposed by Cai and Wang (2008). They used kernel estimator obtained by inverting the weighted double kernel local linear estimate of the conditional distribution function and showed that this estimator is consistent and asymptotically normal distributed as is the tail sample mean. Many other estimations of CVaR can be found in literature. Nevertheless, most authors agree that the tail sample mean is a solid estimator for a sufficiently large sample and moreover it has closed form formula for variance.

2.4 Methods of calculating VaR

There are three methods of calculating VaR: the historical method, the variance-covariance method and the Monte Carlo simulation.

In this chapter, I presented the historical method that simply sorts historical losses in ascending order. Then the Value-at-Risk is chosen as the corresponding sampled quantile. The advantages and the disadvantages of this method are

- + simplicity and wide applicability,
- + no assumption of normal distribution,
- + not necessary to compute correlation coefficients,
- assumption of large dataset,
- the newest data have the same weight as the oldest.

The variance-covariance method assumes that the losses are normally distributed with a zero mean. This is the reason why this method is not often used. On the other hand, it is a very simple method on how to compute Value-at-Risk. It requires only to estimate a standard deviation. Then the VaR is

$$\text{VaR}_\alpha(X) = \hat{\sigma}\phi^{-1}(\alpha),$$

where $\phi^{-1}(\alpha)$ is an α -quantile of standard normal distribution and $\hat{\sigma}$ is the estimated standard deviation. So the advantages and the disadvantages of this method are

- + only one estimated parameter,
- + not time-consuming,
- assumption of normal distribution and zero mean,
- method cannot deal with non-linear profile of a financial portfolio.

Finally, the method of Monte Carlo simulation is more flexible than the previous two methods. It is used in case when we do not have enough history length or in case when we suppose that the future process would be different from the past that is the paramount factor of why Solvency II and Basel capital requirement

calculations predominantly use this method. More information about Monte Carlo methods for Value-at-risk were presented by Glasserman et al. (2000). The advantages and the disadvantages of this method are

- + the most complex method of computing VaR,
- + specification of a stochastic process,
- time and computer consuming,
- different results of consecutive same calculations,
- assumption of depth statistical knowledge.

3. Test statistics

3.1 Tests of VaR accuracy

Tests of VaR accuracy can be divided into 3 sections: unconditional coverage tests, independence tests and joint tests of unconditional coverage and independence.

3.1.1 Unconditional coverage tests for VaR

In this section, I will provide test statistics for the null hypothesis

$$H_0 : \text{VaR}_\alpha(X) = a, \quad a \in \mathbb{R},$$

where α is such that $0 < \alpha < 1$. This hypothesis can be rewritten equivalently as

$$H_0 : 1 - \alpha = P[X \geq a]. \quad (3.1)$$

Let $X_1, \dots, X_n$ be the i.i.d. sample and $Y_1, \dots, Y_n$ be such that $Y_i = \mathbf{1}[X_i \geq a], i = 1, \dots, n$. Hence, $Y_1, \dots, Y_n$ can be considered as i.i.d sample of random variable representing if X_i exceeded $\text{VaR}_\alpha(X)$ or not. This sequence of failures and successes is called Binomial trial and number of failures $x = \sum Y_i$ follows a binomial distribution with probability of exactly k failures given by

$$P[x = k] = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, \dots, n,$$

where p is probability of failure satisfying

$$p = P[Y_i = 1] = P[X_i \geq a] = 1 - \alpha.$$

Assuming that the model is accurate, the failure rate $\frac{x}{n}$ is an unbiased estimator of p . We can rewrite the null hypothesis (3.1) by

$$H_0 : 1 - \alpha (= p) = \frac{x}{n}. \quad (3.2)$$

Then, under the null hypothesis (3.2) and due to De Moivre-Laplace theorem, the test statistic

$$U = \frac{x - np}{\sqrt{np(1 - p)}}$$

follows asymptotically standard normal distribution. We can reject hypothesis (3.2) on the significance level γ when $|U| \geq u(\frac{\gamma}{2})$, where $u(x)$ is a quantile function of the standard normal distribution.

Kupiec POF-test

Another test based on failure rate has been presented by Kupiec (1995). This test is known as POF-test¹ and examines how many times a Value-at-Risk is exceeded over a given time interval. The null hypothesis for the POF-test is

$$H_0 : p = \frac{x}{n},$$

¹proportion of failures

where x is the number of failures. The test statistic is conducted as a LR^2 test and takes the form

$$LR_{POF} = -2\ln \left(\frac{(1-p)^{n-x} p^x}{\left(1 - \frac{x}{n}\right)^{n-x} \left(\frac{x}{n}\right)^x} \right) \quad (3.3)$$

and under the null hypothesis follows asymptotically χ^2 distribution with one degree of freedom.

This test has two shortcomings. First, recognized by Kupiec, is that the test is statistically weak with one year sample sizes. Second, the test takes in account only the number of failures and not the time of occurrence. This is the reason why also independence tests are recommended to be used.

TUFF-test

Another type of backtest, which was suggested by Kupiec (1995), is called TUFF-test³. This test measures time until first failure. If we assume that failures are binomially distributed then the probability of first failure is same as the probability of failure. We define time until first failure in the sample as v . Then the null hypothesis is

$$H_0 : p = \frac{1}{v}.$$

The TUFF-test statistic takes form

$$LR_{TUFF} = -2\ln \left(\frac{p(1-p)^{v-1}}{\left(\frac{1}{v}\right) \left(1 - \frac{1}{v}\right)^{v-1}} \right) \quad (3.4)$$

and under the null hypothesis follows asymptotically χ^2 distribution with one degree of freedom.

TUFF-test ignores the occurrence of failures, just as POF-test does. Furthermore, Dowd (1998) showed that this test has low power in identifying bad VaR models and presented example of model where daily VaR was estimated at 99% confidence level where the sample had 100 observations. The first occurrence of failure was already on the day 7, but the model still wasn't rejected, even if the probability of failure on the day 7 is very low in this model. Due to the severe lack of power, it is recommended to use TUFF-test only when we don't have large data.

3.1.2 Independence tests for VaR

The most known test of independence for VaR is Markov test which was proposed by Christofferssen (1998). This test examines if probability of exceedance of VaR depends on the outcome of the previous observation. Let $Y_1, \dots, Y_n$ have the same meaning as in Section 3.1.1. Then we define numbers

$$n_{kl} = \sum_{i=2}^n \mathbf{1}[Y_{i-1} = k, Y_i = l], \quad k = 0, 1, l = 0, 1,$$

and outcome can be displayed in contingency Table 3.1.

²likelihood-ratio

³Time Until First Failure

	$Y_{i-1} = 0$	$Y_{i-1} = 1$	
$Y_i = 0$	n_{00}	n_{10}	$n_{00} + n_{10}$
$Y_i = 1$	n_{01}	n_{11}	$n_{01} + n_{11}$
	$n_{00} + n_{01}$	$n_{10} + n_{11}$	n

Table 3.1: Contingency table

Additionally, we define probabilities

$$\begin{aligned}\pi_0 &= \frac{n_{01}}{n_{00} + n_{01}}, \\ \pi_1 &= \frac{n_{11}}{n_{10} + n_{11}}, \\ \pi &= \frac{n_{01} + n_{11}}{n_{00} + n_{01} + n_{10} + n_{11}},\end{aligned}$$

where π_0 is the probability of failure where the previous observation wasn't a failure and π_1 is the probability of failure where the previous observation was also a failure. The probability π is unconditional probability of failure. If the model is accurate, then the probability of failure should not depend on previous observation and so the probabilities π_0 and π_1 should be equal. The null hypothesis is

$$H_0 : \pi_0 = \pi_1$$

and the test statistic of Markov test is

$$LR_M = -2\ln \left(\frac{(1 - \pi)^{n_{00} + n_{10}} \pi^{n_{01} + n_{11}}}{(1 - \pi_0)^{n_{00}} \pi_0^{n_{01}} (1 - \pi_1)^{n_{10}} \pi_1^{n_{11}}} \right). \quad (3.5)$$

Under the null hypothesis, the test statistic LR_M follows asymptotically χ^2 distribution with one degree of freedom.

Later, Christoffersen and Pelletier (2004) presented a similar test but compared to Markov test, this test measures also duration between failures. They also provided evidence that this test has higher power against more general forms of dependence, without the need for any additional information.

The similar test was also presented by Haas (2001). He used ideas of Christoffersen and Kupiec and a proposed test which takes in account the duration between failures. We define the duration between failures as v_i , $i = 2, \dots, x$, which is the time between i -th and $(i - 1)$ -th failure and v_1 , which is the time until first failure, same as v in TUFF-test. The null hypothesis is

$$H_0 : \text{Failures are independent from each other.}$$

When x is the number of failures then the test statistic for independence takes the form

$$LR_{IND} = \sum_{i=1}^x \left[-2\ln \left(\frac{p(1 - p)^{v_i - 1}}{\left(\frac{1}{v_i}\right) \left(1 - \frac{1}{v_i}\right)^{v_i - 1}} \right) \right] \quad (3.6)$$

and under the null hypothesis follows asymptotically χ^2 distribution with x degrees of freedom.

3.1.3 Join test of VaR

Join tests combine coverage tests and independence tests and can examine both together. Both properties are required for an accurate VaR measure. One of these tests is called Mixed Kupiec-test, which was presented by Haas (2001). This test combines POF-test for testing of unconditional coverage and test for independence from previous section. The test statistic takes form

$$LR_{MK} = LR_{POF} + LR_{IND} \quad (3.7)$$

and follows asymptotically χ^2 distribution with $x + 1$ degrees of freedom.

3.1.4 Regulatory framework

Basel Committee on Banking Supervision (1996) imposed a capital requirement on banks whose equity trades on the financial markets. They are allowed to use own VaR models which have to be backtested by traffic light coverage test. Basel Committee on Banking Supervision (2006) came with requirement that banks have to compute 10-day VaR on a confidence level 99% in 100 years. As a practical matter, they allowed to compute 1-day VaR on a confidence level 99% and scale it by $\sqrt{10}$.

Backtesting process is based on calculating last 250 daily VaR results and comparing them with daily profits and losses. By the number of failures, the VaR model is classified into one of the three categories as green, yellow or red, as is shown in Table 3.2.

Zone	Number of failures	Cumulative probability
green	0	0.0811
	1	0.2858
	2	0.5432
	3	0.7581
	4	0.8922
yellow	5	0.9588
	6	0.9863
	7	0.9960
	8	0.9989
	9	0.9997
red	10 or more	0.9999

Table 3.2: Traffic light approach defined by Basel Committee on Banking Supervision (1996)

Cumulative probability in Table 3.2 is the probability of obtaining corresponding number of failures when the model is correct, so when the true coverage is 0.99. The same test can be done even for different sample size. Then the lower boundary of yellow zone begins at the point when cumulative probability is higher than 0.95 and red zone begins at point when cumulative probability is higher than 0.9999.

When the result of traffic light test is green, then the model is considered as accurate and it is not necessary to improve it. When the result is yellow, then

the model can be accurate, but also with relatively high probability it can be inaccurate. It is recommended to examine if the VaR model can be improved. When the result is red, there exists only a very small probability that the model is accurate. In this case, the model should be recast.

3.1.5 Testing shortcoming

For tests of coverage we have positively defined null hypotheses and it is not difficult also to define an alternative hypothesis. In case of tests of independence, it is very difficult to define an alternative hypothesis. It can cause any type of dependence in case of rejecting the model, for example clustering. We have defined two tests of independence, but non of them is able to cover all types of dependencies. The same applies for join tests which combine the coverage test and the test of independence. It means that when we reject the model, the reason can be bad coverage, dependences or even both of them. No single test is possible to cover all kinds of problematic properties of the VaR model and when one test does not reject our model, it does not mean that another will not reject it. This is the reason why backtests should be given only as recommendatory instruments when we are finding an accurate model of VaR. We should always try more then one test and analyse the results that we obtain.

3.2 Loss function backtesting of VaR

When we are not only interested in the coverage of VaR and would like to know more about, for example, the magnitude of failures, we will have to use a different framework than I have presented above. This leads to constructing a loss function which can be created by using past data of profits and losses. Lopez (1999) suggested to use quadratic loss function that takes form

$$L(\widehat{\text{VaR}}_{\alpha}^t(Y), Y_t) = \begin{cases} 1 + (Y_t - \widehat{\text{VaR}}_{\alpha}^t(Y))^2, & Y_t \leq \widehat{\text{VaR}}_{\alpha}^t(Y), \\ 0, & Y_t > \widehat{\text{VaR}}_{\alpha}^t(Y), \end{cases}$$

where Y_t is the observed value of profit and loss and $\widehat{\text{VaR}}_{\alpha}^t(Y)$ is estimated VaR with confidence level $\alpha = 0.05$ corresponding to time t . Then the backtest, using this loss function, is based on the sample average loss, defined by formula

$$\widehat{L} = \frac{1}{n} \sum_{t=1}^n \left(L(\widehat{\text{VaR}}_{\alpha}^t(Y), Y_t) \right)$$

and the value of sample average loss is compared to quantile of empirical distribution of simulated average loss. Lopez (1999) suggested a procedure how on to determine the empirical distribution for average loss in three steps:

1. Fit a statistical model to observed profits and losses and estimate suitable distribution of profits and losses.
2. Generate sample, using estimated distribution in first step, and construct a value for the average loss $\widehat{L}_i$.

3. Repeat the second step a sufficient amount of times, for example 200 000 times. The resulting set of average loss $\{\widehat{L}_i\}$ is appropriate empirical distribution of the average loss.

3.3 Test statistic for the location of CVaR

In this section, I will present an asymptotic test for the null hypothesis

$$H_0 : \text{CVaR}_\alpha(X) = c, \quad c \in \mathbb{R}.$$

Before deriving this test I include a Multivariate Delta method which helps to compute a limiting distribution of a function of multivariate normal random vector.

Theorem 3 (Multivariate Delta method).

Let $\boldsymbol{\theta}$ be r -dimensional random vector satisfying for some $\mathbf{b} \in \mathbb{R}^r$ and $\mathbf{V} \in \mathbb{R}^{r \times r}$ that

$$\sqrt{n}(\boldsymbol{\theta} - \mathbf{b}) \xrightarrow{D} N(\mathbf{0}, \mathbf{V}), \quad n \rightarrow \infty. \quad (3.8)$$

Then

$$\sqrt{n}(g(\boldsymbol{\theta}) - g(\mathbf{b})) \xrightarrow{D} N(\mathbf{0}, \nabla g(\mathbf{b})^T \mathbf{V} \nabla g(\mathbf{b})), \quad n \rightarrow \infty,$$

where the function $g : \mathbb{R}^r \rightarrow \mathbb{R}$ is continuously differentiable at $\mathbf{b}$.

Proof. See (Oehlert; 1992). □

Now I can propose the test statistic with standard normal asymptotic distribution.

Theorem 4 (Test statistic for the CVaR).

Let X be a random variable with a finite third moment and the null hypothesis

$$H_0 : \text{CVaR}_\alpha(X) = c, \quad c \in \mathbb{R},$$

be true. Then the test statistic

$$Z = \sqrt{n(1-\alpha)} \frac{\widehat{\text{CVaR}}_\alpha(X) - c}{\sqrt{\frac{\sum_{i=n\alpha+1}^n (X_i - \widehat{\text{CVaR}}_\alpha(X))^2}{n(1-\alpha)} + \alpha \left(\widehat{\text{CVaR}}_\alpha(X) - \widehat{\text{VaR}}_\alpha(X) \right)^2}}$$

follows asymptotically $N(0, 1)$ distribution.

Proof. (Vékás; 2015, pages 9-10)

Firstly, we will summarize assumption of Multivariate Delta method, Theorem 3:

$$\begin{aligned} r &= n(1-\alpha) + 2, \\ \boldsymbol{\theta} &= (X_{(n\alpha)}, X_{(n\alpha+1)}, \dots, \widehat{\text{CVaR}}_\alpha(X)), \\ \mathbf{b} &= (\text{VaR}_\alpha(X), \text{VaR}_{\alpha-\frac{1}{n}}(X), \dots, \text{VaR}_{\frac{n-1}{n}}(X), \text{CVaR}_\alpha(X)), \\ \mathbf{x} &= (x_0, x_1, \dots, x_{n(1-\alpha)}, y), \\ g(\mathbf{x}) &= \frac{\sqrt{1-\alpha}(y-c)}{\sqrt{\frac{\sum_{i=1}^{n(1-\alpha)} x_i^2}{n(1-\alpha)} + \alpha(y-x_0)^2}}. \end{aligned}$$

Assuming that the null hypothesis is true, so following equalities hold:

$$\begin{aligned} g(\mathbf{b}) &= \mathbf{0}, \\ Z &= \sqrt{n}(g(\boldsymbol{\theta}) - g(\mathbf{b})). \end{aligned}$$

Now, it can be easily verified that $g(x)$ is continuously differentiable at $\mathbf{b}$:

$$\begin{aligned} \left. \frac{dg(\mathbf{x})}{dx_i} \right|_{\mathbf{x}=\mathbf{b}} &= 0, \quad i = 0, 1, \dots, n(1-\alpha), \\ \left. \frac{dg(\mathbf{x})}{dy} \right|_{\mathbf{x}=\mathbf{b}} &= \frac{\sqrt{1-\alpha}}{\sqrt{\frac{\sum_{i=n\alpha+1}^n \left(\text{VaR}_{\frac{i-1}{n}}(X) - \text{CVaR}_\alpha(X) \right)^2}{n(1-\alpha)} + \alpha \left(\text{CVaR}_\alpha(X) - \text{VaR}_{\alpha-\frac{1}{n}}(X) \right)^2}}. \end{aligned}$$

The assumption (3.8) holds due to asymptotic properties of $\text{VaR}_\alpha(X)$ and $\text{CVaR}_\alpha(X)$ in Sections 2.1 and 2.2.

Now we will count limiting variance of Z .

$$\begin{aligned} \lim_{n \rightarrow \infty} \text{var}[Z] &= \nabla g(\mathbf{b})^T \mathbf{V} \nabla g(\mathbf{b}) = \left(\left. \frac{dg(\mathbf{x})}{dy} \right|_{\mathbf{x}=\mathbf{b}} \right)^2 \lim_{n \rightarrow \infty} \text{var} \left[\sqrt{n} \widehat{\text{CVaR}}_\alpha(X) \right] = \\ &= \frac{\text{var}[X|X > \text{VaR}_\alpha(X)] + \alpha \left(\text{CVaR}_\alpha(X) - \text{VaR}_\alpha(X) \right)^2}{\frac{1}{n(1-\alpha)} \sum_{i=n\alpha+1}^n \left(\text{VaR}_{\frac{i-1}{n}}(X) - \text{CVaR}_\alpha(X) \right)^2 + \alpha \left(\text{CVaR}_\alpha(X) - \text{VaR}_{\alpha-\frac{1}{n}}(X) \right)^2}. \end{aligned}$$

The first term in the denominator of the fraction above is a Riemann sum

$$\begin{aligned} &\frac{1}{n(1-\alpha)} \lim_{n \rightarrow \infty} \sum_{i=n\alpha+1}^n \left(F^{-1}\left(\frac{i-1}{n}\right) - \text{CVaR}_\alpha(X) \right)^2 = \\ &= \frac{1}{1-\alpha} \int_p^1 \left(F^{-1}(x) - \text{CVaR}_\alpha(X) \right)^2 dx = \\ &= \frac{1}{1-\alpha} \int_{F^{-1}(\alpha)}^\infty (t - \text{CVaR}_\alpha(X))^2 f(t) dt = \\ &= \text{var}[X|X > \text{VaR}_\alpha(X)], \end{aligned}$$

and from continuity of $F^{-1}(\alpha) = \text{VaR}_\alpha(X)$ it follows that

$$\lim_{n \rightarrow \infty} \text{VaR}_{\alpha-\frac{1}{n}}(X) = \text{VaR}_\alpha(X).$$

Formulas above imply that

$$\lim_{n \rightarrow \infty} \text{var}[Z] = 1,$$

so $Z \xrightarrow{D} N(0, 1)$ due to Multivariate Delta method, Theorem 3, and limiting variance of Z . □

4. Backtesting process

In this chapter, I will present the process of backtesting on three different financial instruments: US Dollar to Euro exchange rate, Apple Inc. share price and 10-year US treasury constant maturity nominal rate. All data is obtained from the FRED¹ Economic database (Federal Reserve Bank of St. Louis; 2016). Computations were done using **R** application (see R Development Core Team (2008)) and whole source code can be found in the Backtesting process source code in the Attachment.

All backtests will be calculated over 250 trading days from 7/6/2015 to 6/30/2016. VaR estimates will be calculated using Formula (2.1) with confidence levels $\alpha = 0.99$, $\alpha = 0.95$ and $\alpha = 0.9$. VaR estimates will be based on floating window of 250 days. This means that I will use a floating window of 250 observations from 7/3/2014 to 6/30/2016. These estimates are not the best that I could choose, but for presenting examples of backtests they are sufficient. All tests will be provided on significant level $\gamma = 0.95$. This significant level can be different from VaR confidence level and the reader should be careful not to interchange them.

In Figures 4.1 and 4.2 we can see prices for USD-EUR exchange rate and Apple share price in USD for 500 days previous to 6/30/2016. In Figure 4.3 we can see 10-Year Treasury Constant Maturity Rate in %.

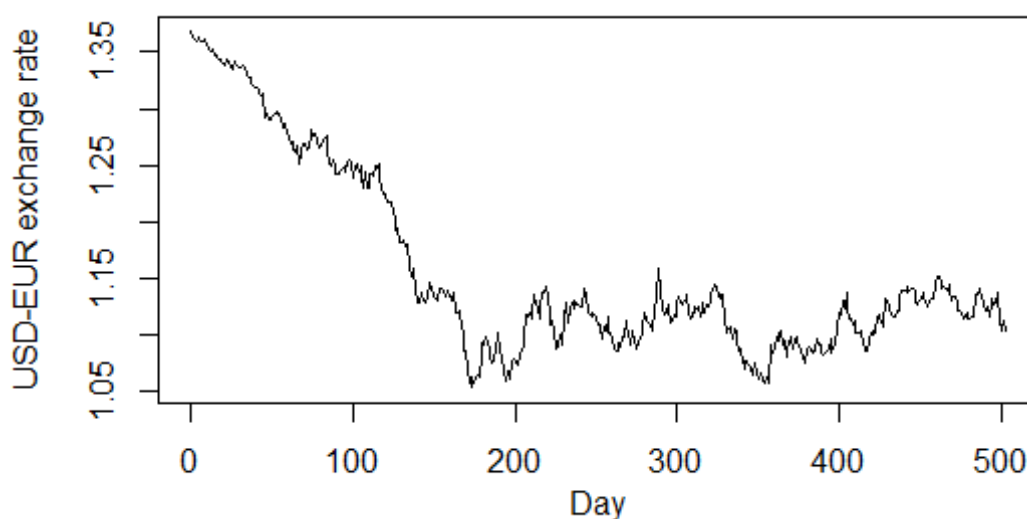


Figure 4.1: USD-EUR exchange rate

¹Federal Reserve Bank of St. Louis Economic Data

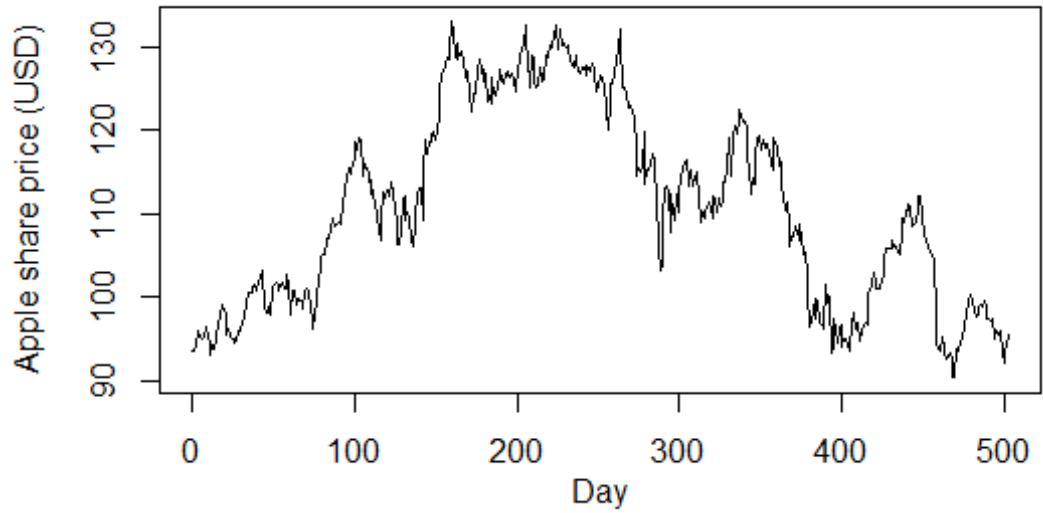


Figure 4.2: Apple share price in USD

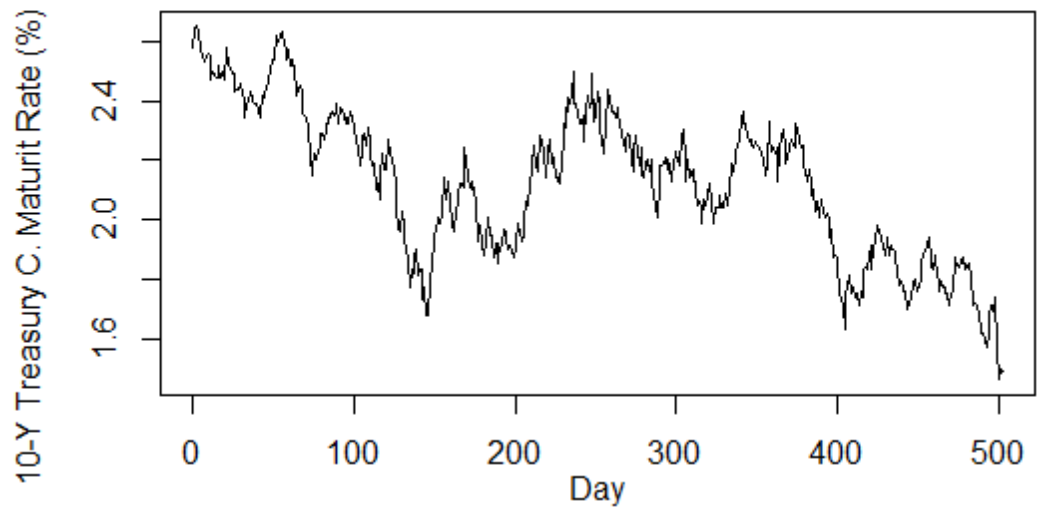


Figure 4.3: 10-Year Treasury Constant Maturity Rate in %

In Table 4.1 we can see descriptive statistics of profits and losses of all three financial instruments.

	USD-EUR	Apple (USD)	Treasury rate (%)
Minimum	-0.0247	-6.8900	-0.1700
1st quartile	-0.0049	-0.8300	-0.0300
Median	-0.0005	0.0000	-0.0100
Mean	-0.0005	0.0041	-0.0022
3rd quartile	0.0034	1.0880	0.0300
Maximum	0.0329	6.1700	0.1500

Table 4.1: Profits and losses descriptive statistics

Profits and losses are displayed in Figures 4.4, 4.5 and 4.6. Estimations of VaR are also displayed in these figures. Blue lines correspond to VaR estimates with confidence level $\alpha = 0.99$, green lines correspond to VaR estimates with confidence level $\alpha = 0.95$ and red lines corresponds to VaR estimates with confidence level $\alpha = 0.9$.

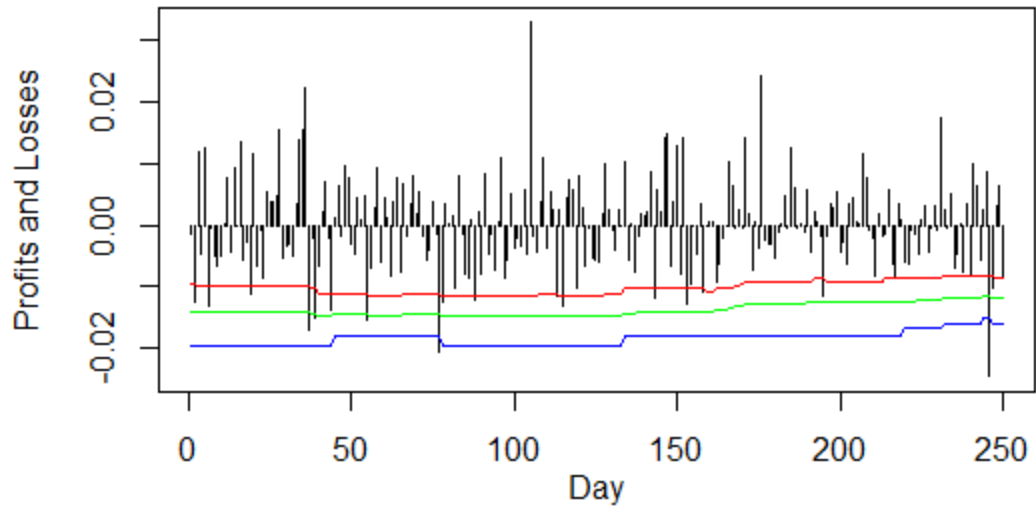


Figure 4.4: Profits and losses and VaR estimates of USD-EUR exchange rate

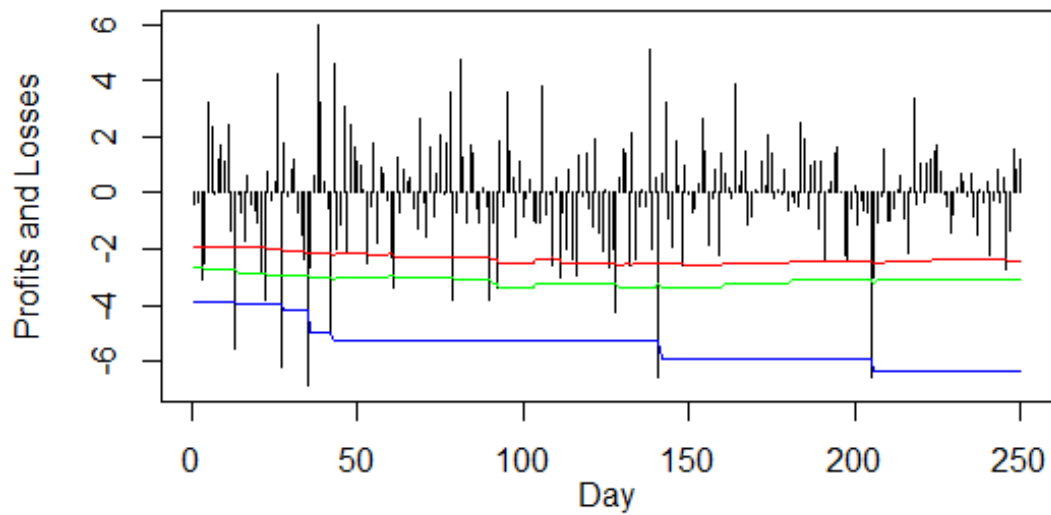


Figure 4.5: Profits and losses and VaR estimates of Apple share price in USD

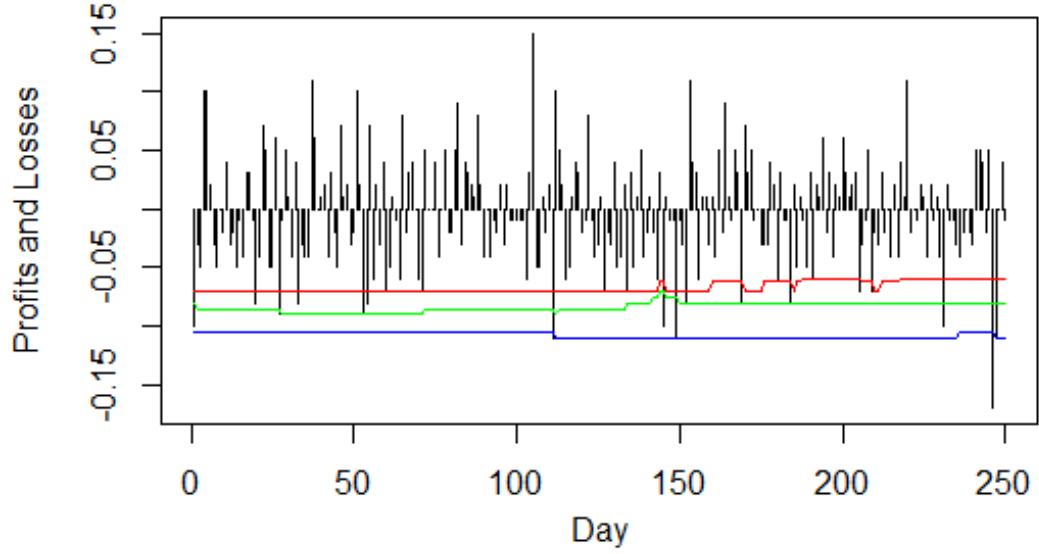


Figure 4.6: Profits and losses and VaR estimates of 10-Year Treasury Constant Maturity Rate in %

Number of failures for each confidence level is the count of black lines that exceed the estimated VaR. These counts are summarized in Table 4.2. The expected numbers are counted by formula

$$(1 - \alpha) * 250,$$

where α is corresponding confidence level of VaR. These numbers are theoretical counts of observations that exceed VaR in an accurate model. We can see that all numbers are close to expected values, only number of failures of VaR with confidence level $\alpha = 0.99$ for Apple shares is a little bit higher than we expected and number of failures of VaR with confidence level $\alpha = 0.95$ for USD-EUR exchange rate is a little bit lower than the expected value.

Conf. level	Expected No.	USD-EUR	Apple	Treasury rate
0.99	2.5	2	6	3
0.95	12.5	5	14	10
0.9	25	19	28	22

Table 4.2: Numbers of failures

First test, I will use, is traffic light coverage test. In Section 3.1.4, I presented zones for confident level $\alpha = 0.99$. Now, I will also need zones for levels $\alpha = 0.95$ and $\alpha = 0.9$. As was mentioned in Section 3.1.4, yellow zone begins at point where cumulative probability is higher than 0.95 and red zone begins at point where cumulative probability is higher then 0.9999. These zones, for all three confident levels, are in Table 4.3.

Zone	0.99	0.95	0.9
Green	0 - 4	0 - 17	0 - 32
Yellow	5 - 9	18 - 26	33 - 43
Red	10 or more	27 or more	44 or more

Table 4.3: Zones for traffic light coverage tests

Results of traffic light coverage test are in Table 4.4. We can see that all results below, beyond the confidence level $\alpha = 0.99$ for Apple shares, are in the green zone. The only one result in yellow zone is the one which I mentioned as problematic one. Nevertheless, the results show that our model seems to be accurate.

	Confidence level	Failures	Result
USD-EUR	0.99	2	green
	0.95	5	green
	0.9	19	green
Apple	0.99	6	yellow
	0.95	14	green
	0.9	28	green
Treasury rate	0.99	3	green
	0.95	10	green
	0.9	22	green

Table 4.4: Zones for traffic light coverage tests

For the next test, I can use is Kupiec POF-test that was presented in Section 3.1.1. This test is test of coverage as traffic light test. It is also based only on number of failures. The value of POF-test statistic is compute using Formula (3.3). This value is compared to critical value $\chi_1^2(0.95)$ that is 0.95-quantile of χ^2 distribution with one degree of freedom. When the value of test statistic is higher than $\chi_1^2(0.95)$, we can reject the model. Otherwise, the model is accepted. The results of POF-tests are in Table 4.5. We can see that nearly in all cases our model was accepted. Only one test was rejected. This is the case which I mentioned as problematic.

	VaR Confidence level	POF-test	$\chi_1^2(0.95)$	Result
USD-EUR	0.99	0.1084	3.8415	Accepted
	0.95	6.0715	3.8415	Rejected
	0.9	1.7299	3.8415	Accepted
Apple	0.99	3.5554	3.8415	Accepted
	0.95	0.1827	3.8415	Accepted
	0.9	0.3866	3.8415	Accepted
Treasury rate	0.99	0.0949	3.8415	Accepted
	0.95	0.5633	3.8415	Accepted
	0.9	0.4152	3.8415	Accepted

Table 4.5: POF-test results

Last test of coverage of VaR is TUFF-test. This test examines whether the time of the first failure is in a correspondence with the confidence level of VaR. It means that in the true model with confidence level $\alpha = 0.99$ the failure should occur only once in 100 days and the first occurrence should be on the day 100. Table 4.6 contains time until the first failure.

	Confidence level	Time until first failute
USD-EUR	0.99	77
	0.95	37
	0.9	2
Apple	0.99	13
	0.95	3
	0.9	3
Treasury rate	0.99	111
	0.95	1
	0.9	1

Table 4.6: Time until first failure

TUFF-test statistic (3.4) is asymptotically χ^2 distributed with one degree of freedom. It means that the critical value is $\chi_1^2(0.95)$ for tests on significant level $\gamma = 0.95$. Results of this test are in Table 4.7. We can see that our model was rejected only for Treasury rate for confidence levels $\alpha = 0.95$ and $\alpha = 0.9$, where the first occurrence of failure was in the first day. We can notice that the model wasn't rejected in case of Apple shares for confidence level $\alpha = 0.95$ where the first failure was in the 3rd day. It only goes to show that this test has low power as I mentioned in Section 3.1.1.

	VaR Confidence level	TUFF-test	$\chi_1^2(0.95)$	Result
USD-EUR	0.99	0.0634	3.8415	Accepted
	0.95	0.4900	3.8415	Accepted
	0.9	2.0433	3.8415	Accepted
Apple	0.99	2.4006	3.8415	Accepted
	0.95	2.3776	3.8415	Accepted
	0.9	1.2075	3.8415	Accepted
Treasury rate	0.99	0.0114	3.8415	Accepted
	0.95	5.9915	3.8415	Rejected
	0.9	4.6052	3.8415	Rejected

Table 4.7: TUFF-test results

Now, I will use Markov test statistic (3.5) to show if failures are independent or not. For this test, I need to count numbers n_{00} , n_{10} , n_{01} and n_{11} and probabilities π_0 , π_1 and π defined in Section 3.1.2. All these numbers are in Table 4.8.

	USD-EUR				Apple		Treasury rate		
	0.99	0.95	0.9	0.99	0.95	0.9	0.99	0.95	0.9
n_{00}	246	240	215	238	223	200	244	231	208
n_{10}	2	5	16	6	13	22	3	9	20
n_{01}	2	5	17	6	13	22	3	9	20
n_{11}	0	0	2	0	1	6	0	1	2
π_0	.0081	.0204	.0733	.0246	.0551	.0991	.0121	.0375	.0877
π_1	0	0	.1111	0	.0714	.2143	0	.1000	.0909
π	.0080	.0200	.0760	.0240	.0560	.1120	.0120	.0400	.0880

Table 4.8: Numbers for Markov test

Markov test statistic follows asymptotically χ^2 distribution with one degree of freedom. The critical value will be same as for POF and TUFF-test. Results of Markov tests are in Table 4.9. This test did not reject any model.

	VaR Conf. level	Markov test	$\chi_1^2(0.95)$	Result
USD-EUR	0.99	0.0323	3.8415	Accepted
	0.95	0.2041	3.8415	Accepted
	0.9	0.3043	3.8415	Accepted
Apple	0.99	0.2951	3.8415	Accepted
	0.95	0.0620	3.8415	Accepted
	0.9	2.7858	3.8415	Accepted
Treasury rate	0.99	0.0729	3.8415	Accepted
	0.95	0.7108	3.8415	Accepted
	0.9	0.0025	3.8415	Accepted

Table 4.9: Markov test results

Another test of independence of VaR failures is based on the test statistic for independence (3.6). It follows asymptotically χ^2 distribution with numbers of freedom equal to number of failures. Critics values on significant level $\gamma = 0.95$ for all tests are showed in Table 4.10.

Degrees of freedom	0.95-quantile
2	5.9915
3	7.8147
5	11.0705
6	12.5916
10	18.3070
14	23.6848
19	30.1435
22	33.9244
28	41.3371

Table 4.10: 0.95-quantiles of χ^2 distribution with different degrees of freedom

Test of independence examines durations between failures. For my data, all durations are in Table 4.11.

Order	USD-EUR			Apple			Treasury rate		
	0.99	0.95	0.9	0.99	0.95	0.9	0.99	0.95	0.9
1	77	37	2	13	3	3	111	1	1
2	169	2	4	14	10	1	38	26	19
3		16	13	8	8	9	97	26	7
4		22	18	7	1	8		58	6
5		169	2	99	5	1		34	20
6			5	64	8	5		4	1
7			11		7	7		3	6
8			22		19	1		79	11
9			1		18	1		15	40
10			10		11	6		1	16
11			25		2	11			7
12			2		36	7			11
13			28		13	1			4
14			10		64	18			3
15			5			11			17
16			37			2			15
17			51			17			7
18			1			2			14
19			3			5			4
20						10			22
21						2			15
22						4			1
23						9			
24						7			
25						43			
26						14			
27						1			
28						40			

Table 4.11: Durations since previous failures

In Table 4.12 we can see results of the tests of independence. We can see that more than half of tests were rejected. This implies that failures are not independent and our model could be inaccurate.

	VaR Conf. level	Indep. test	$\chi_x^2(0.95)$	Result
USD-EUR	0.99	0.3968	5.9915	Accepted
	0.95	14.8424	11.0705	Rejected
	0.9	30.8509	30.1435	Rejected
Apple	0.99	11.7541	11.0705	Rejected
	0.95	18.8255	23.6858	Accepted
	0.9	45.7278	41.3371	Rejected
Treasury rate	0.99	0.7178	7.8147	Accepted
	0.95	21.7548	18.3070	Rejected
	0.9	24.2143	33.9244	Accepted

Table 4.12: Independence test results

Lastly, I will use join test (3.7). The test statistic is sum of POF-test and test of independence and follows asymptotically χ^2 distribution with degrees of freedom equal to number of failures increased by one. Results of tests are in Table 4.13. We can see that results are the same as in the previous test.

	VaR Conf. level	Join test	$\chi_{x+1}^2(0.95)$	Result
USD-EUR	0.99	0.5052	7.8147	Accepted
	0.95	20.9139	12.5916	Rejected
	0.9	32.5897	31.4104	Rejected
Apple	0.99	15.3095	14.0671	Rejected
	0.95	19.0082	24.9958	Accepted
	0.9	46.1144	42.5569	Rejected
Treasury rate	0.99	0.8127	9.4877	Accepted
	0.95	22.3182	19.6751	Rejected
	0.9	24.6295	35.1725	Accepted

Table 4.13: Join test results

	TLCT ²	POF	TUFF	Markov	Indep.	Join
USD-EUR						
0.99	G	A	A	A	A	A
0.95	G	R	A	A	R	R
0.9	G	A	A	A	R	R
Apple						
0.99	Y	A	A	A	R	R
0.95	G	A	A	A	A	A
0.9	G	A	A	A	R	R
Treasury rate						
0.99	G	A	A	A	A	A
0.95	G	A	R	A	R	R
0.9	G	A	R	A	A	A

G = green, Y = yellow, A = accepted, R = rejected

Table 4.14: Summary of all tests

Finally, we can see all results in Table 4.14. Based on all tests, I can undoubtedly recommend to use my model for USD-EUR exchange rate and 10-Year treasury constant maturity rate, when we calculate VaR estimates with confidence level $\alpha = 0.99$ and for Apple shares, when VaR estimates are calculated with confidence level $\alpha = 0.95$. In these cases, I didn't reject the model for all tests and also the Traffic light test was within the green zone.

In the case of 10-Year Treasury constant maturity rate, the VaR model with confidence level $\alpha = 0.9$ was rejected only once, based on TUFF-test. All other tests did not reject it, so we can say that even in this case the model is accurate.

For USD-EUR exchange rate and 10-Year treasury constant maturity rate VaR model with confidence level $\alpha = 0.95$ and for Apple shares VaR model with confidence level $\alpha = 0.95$, the model was rejected three times out of six tests. We should think about that and try to improve the model.

5. Numerical experiments

In this chapter, I would like to show statistical power and probability of Type I error of selected tests of VaR coverage and independence. I have chosen POF-test, TUFF-test and test of independence. I assumed probably the three most commonly used distributions of returns or profits and losses. These distributions are:

- Standard normal distribution¹,
- Pareto distribution² with scale e and shape 3,
- Gamma distribution³ with scale 1 and shape 2.

5.1 Statistical power

For this experiment, I generated 1000 samples of size $n = 1000$ observations for each distribution. Then, I performed all tests using 40 different theoretical VaRs close to the true value of VaR with confident level $\alpha = 0.95$. Values of true VaR are in Table 5.1. All computations were provided using **R** application (see (R Development Core Team; 2008)) and the whole source code can be found in the Statistical power source code in Attachment.

	VaR _{0.95} (X)
Standard normal distribution	2.3263
Pareto distribution	16.3260
Gamma distribution	6.6384

Table 5.1: True values of VaR with confidence level $\alpha = 0.95$

In Figures 5.1, 5.2 and 5.3, we can see results of the experiment. We can see, that for all distributions, the power is lowest for TUFF-test and highest for POF-test. This means that the TUFF test has very low power in identifying bad VaR models. From these results, we can say that POF-test is the best in identifying bad VaR models.

¹pdf of Standard normal distribution is $f(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2\sigma^2}}$, $x \in \mathbb{R}$

²pdf of Pareto distribution with scale s and shape α is $f(x) = \frac{\alpha s}{x^{\alpha+1}}$, $x \in [s; \infty)$

³pdf of Gamma distribution with scale ϕ and shape a is $f(x) = \frac{1}{\Gamma(a)\phi^a}x^{a-1}e^{-\frac{x}{\phi}}$, $x \in \mathbb{R}$

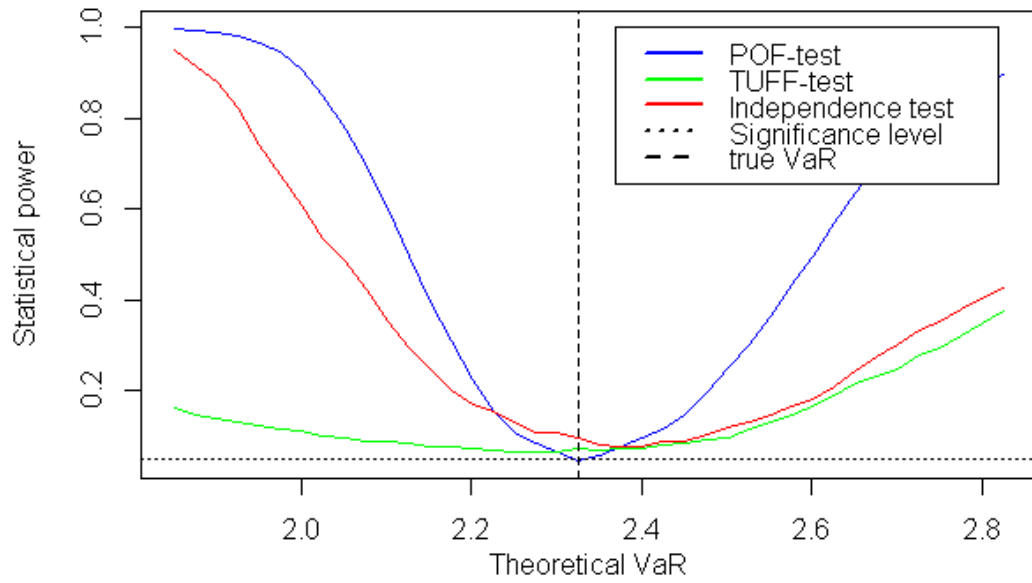


Figure 5.1: Statistic power for Standard normal distribution and 3 different tests

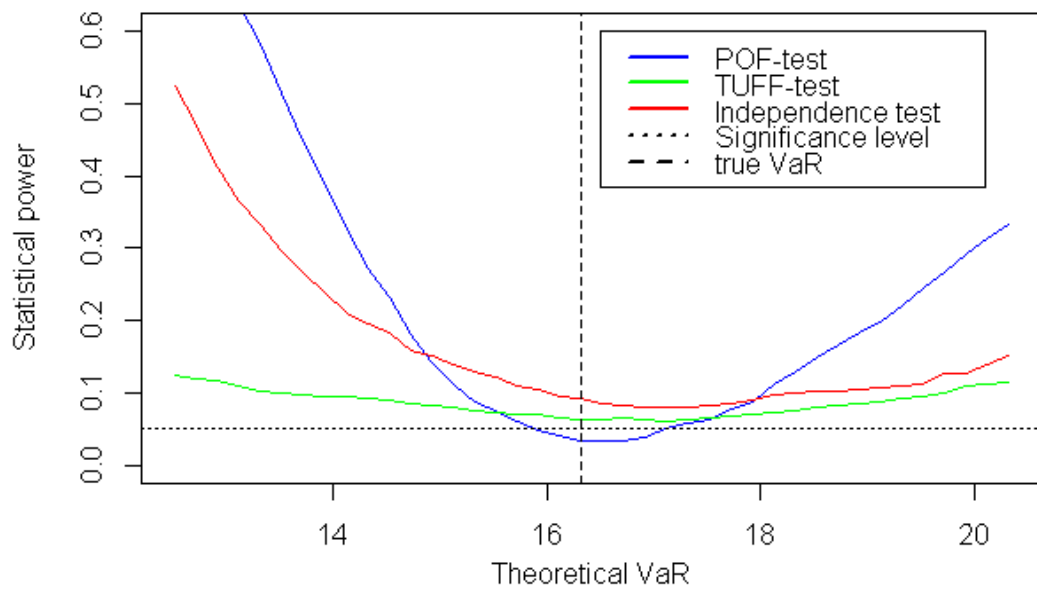


Figure 5.2: Statistic power for Pareto distribution and 3 different tests

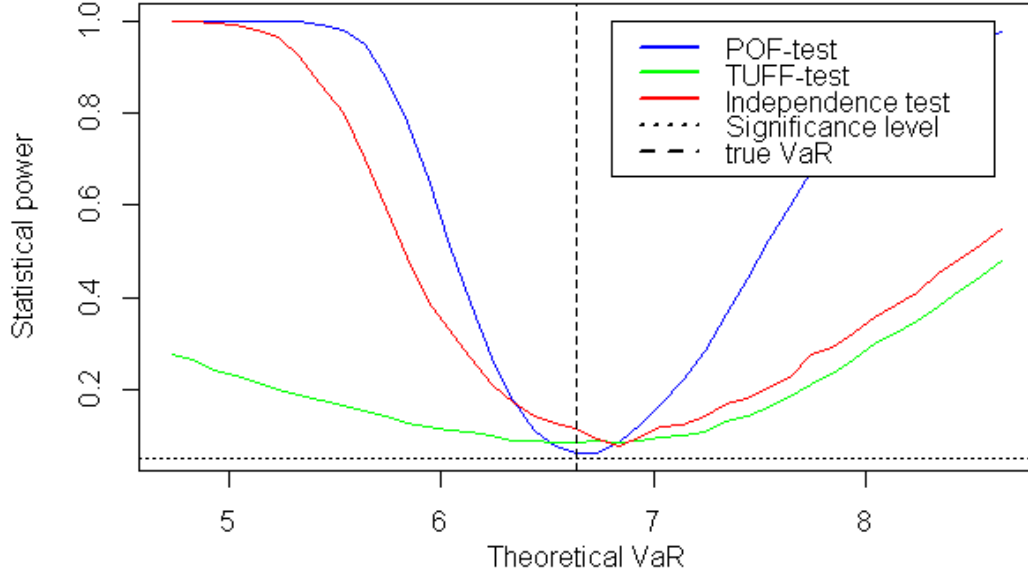


Figure 5.3: Statistic power for Gamma distribution and 3 different tests

5.2 Probability of Type I error

In this experiment, I generated 1000 times samples with size 200, 250, $\dots$, 2000 observations. For each sample size I calculated probability of Type I error on significant level $\gamma = 0.95$. All tests were provided to test VaR with confidence level $\alpha = 0.95$. Results are plotted in Figures 5.4, 5.5 and 5.6. All computations were provided using **R** application (see (R Development Core Team; 2008)) and the whole source code can be found in the Probability of Type I error source code in the Attachment.

We can see, that for all three distributions, sufficient sample size starts at approximately 500 observations. From this count the probability of Type I error doesn't rapidly change.

The lowest probability of Type I error is for POF-test. For sample sizes 500 and more it fluctuates around 0.05. The probability for TUFF-test is approximately 0.07, for sample sizes larger then 500 as well.

We can also notice, that the probability of Type I error for test of independence grows up for samples with size from 500. We should have it in mind when we test VaR models for large samples.



Figure 5.4: Probability of Type I error for Standard normal distribution and 3 different tests



Figure 5.5: Probability of Type I error for Pareto distribution and 3 different tests



Figure 5.6: Probability of Type I error for Gamma distribution and 3 different tests

Conclusion

Using empirical data, I showed how the process of backtesting of Value at Risk model can be done. We could observe that even if one test rejected the model, the other test accepted it. The reason why this happens is that each test is testing different properties of Value at Risk. Because of that we should not have built a conclusion based only on one test. We should do as many tests as we can and then compare and discuss the results. One rejection does not have to mean that our model is inaccurate.

We have to also take into account that tests of accuracy of VaR depend on the sample size. In last Chapter 5, I showed that for POF-test, TUFF-test and independence test, the sample size larger than 500 observations should be taken into account. In all cases, the probability of Type I error is increasing rapidly when sample size is smaller than 500.

I also showed that TUFF-test has very low statistical power in comparison to POF-test and test of independence, and therefore we should not give them such weight in deciding whether to reject the model.

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Attachment

Backtesting process source code

```
# loading libraries
library(ES)
library(PerformanceAnalytics)
library(VGAM)
set.seed(20160101)

# log normal density with VaR and CVaR
grid = seq(0,25,.1)
dataa = dlnorm(grid,1,0.5)
plot(grid,dataa,type="l",xlab="x",ylab="f(x)", col="blue")
legend("top",c("Density of log-normal distribution"),lty=1,col="blue",
> cex=0.8)
VaR <- qlnorm(0.95,1,0.5)
CVaR <- qlnorm(mean(dataa[238:251]),1,0.5,lower.tail = FALSE)

#loading data
setwd("C:/DATA/Škola/mgr/Diplomka/")
data = read.csv("dip.csv")
data$dif = c()

#basic plots and information about EUR/USD
data$dif_eur_usd = c(0, diff(data$eur_usd))
plot(data$eur_usd, type="l", xlab="Day", ylab="USD-EUR exchange rate")
plot(data$dif_eur_usd, type="h", xlab="Day",
> ylab="USD-EUR exchange rate")
summary(data$dif_eur_usd)
hist(data$dif_eur_usd)

#basic plots and information about Apple shares
data$dif_apple = c(0, diff(data$apple))
plot(data$apple, type="l", xlab="Day", ylab="Apple share price (USD)")
plot(data$dif_apple, type="h", xlab="Day",
> ylab="Apple share price (USD)")
summary(data$dif_apple)
hist(data$dif_apple)

#basic plots and information about 10-Y bonds
data$dif_bond = c(0, diff(data$bond))
plot(data$bond, type="l", xlab="Day",
> ylab="10-Y Treasury C. Maturit Rate (%)")
plot(data$dif_bond, type="h", ylab="10-Y Treasury C. Maturit Rate (%)")
summary(data$dif_bond)
hist(data$dif_bond)

#zones for traffic light coverage test for alpha=0.99
yellow = 0
```

```

red = 0
for (i in 0:11) {
  print(i)
  temp_prob = pbinom(i, size=250, prob=0.01)
  print(temp_prob)
  if (temp_prob < 0.95) {
    yellow = i + 1
  }
  if (temp_prob < 0.9999) {
    red = i + 1
  }
}
yellow
red

#zones for traffic light coverage test for alpha=0.95
yellow = 0
red = 0
for (i in 0:30) {
  print(i)
  temp_prob = pbinom(i, size=250, prob=0.05)
  print(temp_prob)
  if (temp_prob < 0.95) {
    yellow = i + 1
  }
  if (temp_prob < 0.9999) {
    red = i + 1
  }
}
yellow
red

#zones for traffic light coverage test for alpha=0.9
yellow = 0
red = 0
for (i in 0:60) {
  print(i)
  temp_prob = pbinom(i, size=250, prob=0.1)
  print(temp_prob)
  if (temp_prob < 0.95) {
    yellow = i + 1
  }
  if (temp_prob < 0.9999) {
    red = i + 1
  }
}
yellow
red

#####
#                               backtesting                               #

```

```

#following code have to be run three times.#
#      1. with data$dif_eur_usd      #
#      2. with data$dif_apple        #
#      3. with data$dif_bond         #
#####
#vectors of VaR estimations
data$var99 = c(0:0)
data$var95 = c(0:0)
data$var90 = c(0:0)

#vectors of 0 and 1, where 1 means, that loss is higher
#than estimation of VaR
data$var99_fail = c(0:0)
data$var95_fail = c(0:0)
data$var90_fail = c(0:0)

#computation of VaR estimations based on last 250 daily profits
#and losses
for (i in 1:250) {
  temp = data$dif_eur_usd[(i+2):(i+251)]
  data$var99[i+252] = quantile(temp, probs=0.01)
  data$var95[i+252] = quantile(temp, probs=0.05)
  data$var90[i+252] = quantile(temp, probs=0.10)
}

#plot of profits and losses and VaR estimations
plot(data$dif_eur_usd[253:502], type='h',xlab="Day",
> ylab="Profits and Losses")
lines(data$var99[253:502], col="blue")
lines(data$var95[253:502], col="green")
lines(data$var90[253:502], col="red")

#number of failures and times of failures
var99_fail = 0
var95_fail = 0
var90_fail = 0
v_99 = c()
v_95 = c()
v_90 = c()

#counting number of failures and times of failures
for (i in 253:502) {
  if (data$dif_eur_usd[i]<=data$var99[i]) {
    var99_fail = var99_fail + 1
    v_99 = c(v_99, i-252)
    data$var99_fail[i] = 1
  }

  if (data$dif_eur_usd[i]<=data$var95[i]) {
    var95_fail = var95_fail + 1
    v_95 = c(v_95, i-252)
  }
}

```

```

    data$var95_fail[i] = 1
  }

  if (data$dif_eur_usd[i]<=data$var90[i]) {
    var90_fail = var90_fail + 1
    v_90 = c(v_90, i-252)
    data$var90_fail[i] = 1
  }
}
var99_fail
var95_fail
var90_fail
v_99
v_95
v_90

#0.95-quantile of chi-squared distribution
chi2_95 = qchisq(.95, df=1)

#POF-tests
LR_POF_99 = -2*log(((1-0.01)^(250-var99_fail)*0.01^var99_fail)/
> ((1-(var99_fail/250))^(250-var99_fail)*(var99_fail/250)^
> (var99_fail)))
LR_POF_95 = -2*log(((1-0.05)^(250-var95_fail)*0.05^var95_fail)/
> ((1-(var95_fail/250))^(250-var95_fail)*(var95_fail/250)^
> (var95_fail)))
LR_POF_90 = -2*log(((1-0.10)^(250-var90_fail)*0.10^var90_fail)/
> ((1-(var90_fail/250))^(250-var90_fail)*(var90_fail/250)^
> (var90_fail)))

#TUFF-tests
LR_TUFF_99 = -2*log(((1-0.01)^(v_99[1]-1)*0.01)/((1-(1/v_99[1]))^
> (v_99[1]-1)*(1/v_99[1])))
LR_TUFF_95 = -2*log(((1-0.05)^(v_95[1]-1)*0.05)/((1-(1/v_95[1]))^
> (v_95[1]-1)*(1/v_95[1])))
LR_TUFF_90 = -2*log(((1-0.10)^(v_90[1]-1)*0.10)/((1-(1/v_90[1]))^
> (v_90[1]-1)*(1/v_90[1])))

#Markov tests' numbers
n_00_99 = 0
n_10_99 = 0
n_01_99 = 0
n_11_99 = 0
n_00_95 = 0
n_10_95 = 0
n_01_95 = 0
n_11_95 = 0
n_00_90 = 0
n_10_90 = 0
n_01_90 = 0
n_11_90 = 0

```

```

#counting Markov tests' numbers
for (i in 253:502) {
  if ((data$var99_fail[i]==1)&&(data$var99_fail[i-1]==1)) {
    n_11_99 = n_11_99 + 1
  }
  if ((data$var99_fail[i]==1)&&(data$var99_fail[i-1]==0)) {
    n_10_99 = n_10_99 + 1
  }
  if ((data$var99_fail[i]==0)&&(data$var99_fail[i-1]==1)) {
    n_01_99 = n_01_99 + 1
  }
  if ((data$var99_fail[i]==0)&&(data$var99_fail[i-1]==0)) {
    n_00_99 = n_00_99 + 1
  }

  if ((data$var95_fail[i]==1)&&(data$var95_fail[i-1]==1)) {
    n_11_95 = n_11_95 + 1
  }
  if ((data$var95_fail[i]==1)&&(data$var95_fail[i-1]==0)) {
    n_10_95 = n_10_95 + 1
  }
  if ((data$var95_fail[i]==0)&&(data$var95_fail[i-1]==1)) {
    n_01_95 = n_01_95 + 1
  }
  if ((data$var95_fail[i]==0)&&(data$var95_fail[i-1]==0)) {
    n_00_95 = n_00_95 + 1
  }

  if ((data$var90_fail[i]==1)&&(data$var90_fail[i-1]==1)) {
    n_11_90 = n_11_90 + 1
  }
  if ((data$var90_fail[i]==1)&&(data$var90_fail[i-1]==0)) {
    n_01_90 = n_01_90 + 1
  }
  if ((data$var90_fail[i]==0)&&(data$var90_fail[i-1]==1)) {
    n_10_90 = n_10_90 + 1
  }
  if ((data$var90_fail[i]==0)&&(data$var90_fail[i-1]==0)) {
    n_00_90 = n_00_90 + 1
  }
}

#counting Markov tests' probabilities
pi_99 = (n_01_99+n_11_99)/(n_00_99+n_01_99+n_10_99+n_11_99)
pi_0_99 = n_01_99/(n_00_99+n_01_99)
pi_1_99 = n_11_99/(n_10_99+n_11_99)
pi_95 = (n_01_95+n_11_95)/(n_00_95+n_01_95+n_10_95+n_11_95)
pi_0_95 = n_01_95/(n_00_95+n_01_95)
pi_1_95 = n_11_95/(n_10_95+n_11_95)
pi_90 = (n_01_90+n_11_90)/(n_00_90+n_01_90+n_10_90+n_11_90)

```

```

pi_0_90 = n_01_90/(n_00_90+n_01_90)
pi_1_90 = n_11_90/(n_10_90+n_11_90)

#Markov tests
LR_M_99 = -2*log(((1-pi_99)^(n_00_99+n_10_99)*pi_99^(n_01_99+n_11_99))
> /((1-pi_0_99)^n_00_99*pi_0_99^n_01_99*(1-pi_1_99)^
> n_10_99*pi_1_99^n_11_99))
LR_M_95 = -2*log(((1-pi_95)^(n_00_95+n_10_95)*pi_95^(n_01_95+n_11_95))
> /((1-pi_0_95)^n_00_95*pi_0_95^n_01_95*(1-pi_1_95)^
> n_10_95*pi_1_95^n_11_95))
LR_M_90 = -2*log(((1-pi_90)^(n_00_90+n_10_90)*pi_90^(n_01_90+n_11_90))
> /((1-pi_0_90)^n_00_90*pi_0_90^n_01_90*(1-pi_1_90)^
> n_10_90*pi_1_90^n_11_90))

#times between failures
v_99_dif = c(v_99[1], diff(v_99))
v_95_dif = c(v_95[1], diff(v_95))
v_90_dif = c(v_90[1], diff(v_90))

#independence tests
LR_ind_99 = 0
for (i in 1:var99_fail) {
  LR_ind_99 = LR_ind_99 - 2*log((0.01*(1-0.01)^(v_99_dif[i]-1))/
  > ((1/v_99_dif[i])*(1-(1/v_99_dif[i]))^(v_99_dif[i]-1)))
}
chi2_95_99 = qchisq(.95, df=var99_fail)

LR_ind_95 = 0
for (i in 1:var95_fail) {
  LR_ind_95 = LR_ind_95 - 2*log((0.05*(1-0.05)^(v_95_dif[i]-1))/
  > ((1/v_95_dif[i])*(1-(1/v_95_dif[i]))^(v_95_dif[i]-1)))
}
chi2_95_95 = qchisq(.95, df=var95_fail)

LR_ind_90 = 0
for (i in 1:var90_fail) {
  LR_ind_90 = LR_ind_90 - 2*log((0.1*(1-0.1)^(v_90_dif[i]-1))/
  > ((1/v_90_dif[i])*(1-(1/v_90_dif[i]))^(v_90_dif[i]-1)))
}
chi2_95_90 = qchisq(.95, df=var90_fail)

#join tests
LR_MK_99 = LR_POF_99 + LR_ind_99
chi2_95_99_1 = qchisq(.95, df=(var99_fail+1))

LR_MK_95 = LR_POF_95 + LR_ind_95
chi2_95_95_1 = qchisq(.95, df=(var95_fail+1))

LR_MK_90 = LR_POF_90 + LR_ind_90
chi2_95_90_1 = qchisq(.95, df=(var90_fail+1))

```


Statistical power source code

```
#####  
#following code have to be run three times.      #  
#      1. quantile = qnorm(0.99)-0.5+k*0.025      #  
#      temp = rnorm(sample_size)                  #  
#      2. quantile = qpareto(0.99, 3, exp(1),      #  
#      > lower.tail = TRUE)-4+k*0.2              #  
#      temp = rpareto(sample_size, 3, exp(1))      #  
#      3. quantile = qgamma(0.99, 2)-2+k*0.1        #  
#      temp = rgamma(1000, 2)                      #  
#####  
  
#initialization of vectors  
dobre = c()  
dobre_TUFF = c()  
dobre_ind = c()  
var = c()  
  
for (k in 1:40) {  
  set.seed(20160101)  
  print(k)  
  accurate_POF = 0  
  accurate_TUFF = 0  
  accurate_ind = 0  
  # set the theoretical VaR  
  quantile = qnorm(0.99)-0.5+k*0.025  
  sample_size = 1000  
  p = 0.01  
  var=c(var,quantile)  
  opakovani = 1000  
  
  for (j in 1:opakovani) {  
    temp = rnorm(sample_size)  
    fail = 0  
    vv=c()  
    var_fail = c(0:0)  
    for(i in 1:sample_size) {  
      if (temp[i] > quantile) {  
        fail = fail + 1;  
        vv = c(vv,i)  
        var_fail[i] = 1  
      } else {  
        var_fail[i] = 0  
      }  
    }  
    chi2_95 = qchisq(.95, df=1)  
    ##### POF-test #####  
    LR_POF = -2*log(((1-p)^(sample_size-fail)*p^fail)/((1-(fail  
    > /sample_size))^(sample_size-fail)*(fail/sample_size)^(fail)))  
    ##### TUFF-test #####
```

```

if (is.null(vv)) {
  LR_TUFF = 100
} else {
  LR_TUFF = -2*log(((1-p)^(vv[1]-1)*p)/((1-(1/vv[1]))^(vv[1]-1)
    > *(1/vv[1])))
}

##### independent-test #####
v_dif = c(vv[1], diff(vv))
LR_ind = 0
for (i in 1:fail) {
  LR_ind = LR_ind - 2*log((0.01*(1-0.01)^(v_dif[i]-1))/
    > ((1/v_dif[i])*(1-(1/v_dif[i]))^(v_dif[i]-1)))
}
chi2_95_ex = qchisq(.95, df=fail)

if (fail==0) {
  LR_ind = 1
  chi2_95_ex = 0
}

if (LR_POF <= chi2_95) {
  accurate_POF = accurate_POF + 1;
}
if (LR_TUFF <= chi2_95) {
  accurate_TUFF = accurate_TUFF + 1;
}
if (LR_ind <= chi2_95_ex) {
  accurate_ind = accurate_ind + 1;
}
}

dobre = c(dobre, (opakovani-accurate_POF)/opakovani)
dobre_TUFF = c(dobre_TUFF, (opakovani-accurate_TUFF)/opakovani)
dobre_ind = c(dobre_ind, (opakovani-accurate_ind)/opakovani)
}

# plot of results
plot(var, dobre, ylab="Statistical power", xlab="Theoretical VaR",
  > type="l", col="blue")
lines(var,dobre_TUFF, col="green")
lines(var,dobre_ind, col="red", type="l")
abline(h =0.05, lty=3)
abline(v = qnorm(0.99), lty=2)
legend(2.37,1, lty=c(1,1,1,3,2), c("POF-test", "TUFF-test",
  > "Independence test", "Significance level", "true VaR"),
  > lwd=c(2.5,2.5,2.5,2.5,2.5),col=c("blue", "green",
  > "red", "black", "black"))

```

Probability of Type I error source code

```
#####
#following code have to be run three times.      #
#      1. quantile = qnorm(0.99)                  #
#      temp = rnorm(sample_size)                  #
#      2. quantile = qpareto(0.99, 3, exp(1),      #
#      > lower.tail = TRUE)                        #
#      temp = rpareto(sample_size, 3, exp(1))     #
#      3. quantile = qgamma(0.99, 2)              #
#      temp = rgamma(sample_size, 2)              #
#####
spatne = c()
spatne2 = c()
spatne3 = c()
sample=c()
for (k in 4:40) {
  set.seed(20160101)
  print(k)
  inaccurate = 0
  inaccurate_TUFF = 0
  inaccurate_ind = 0
  quantile = qnorm(0.99)
  sample_size = 50*k
  p = 0.01
  sample=c(sample,50*k)

  for (j in 1:1000) {
    temp = rnorm(sample_size)
    fail = 0
    vv=c()
    var_fail = c(0:0)

    for(i in 1:sample_size) {
      if (temp[i] > quantile) {
        fail = fail +1;
        vv = c(vv,i)
        var_fail[i] = 1
      } else {
        var_fail[i] = 0
      }
    }
  }

  chi2_95 = qchisq(.95, df=1)

  ##### POF-test #####
  LR_POF = -2*log(((1-p)^(sample_size-fail)*p^fail)/((1-(fail/
  > sample_size))^(sample_size-fail)*(fail/sample_size)^(fail)))
  ##### TUFF-test #####
  if (is.null(vv)) {
```

```

    LR_TUFF = 100
  } else {
    LR_TUFF = -2*log(((1-p)^(vv[1]-1)*p)/((1-(1/vv[1]))^(vv[1]-1)*
    > (1/vv[1])))
  }
  ##### independent-test #####
  v_dif = c(vv[1], diff(vv))
  LR_ind = 0
  for (i in 1:fail) {
    LR_ind = LR_ind - 2*log((0.01*(1-0.01)^(v_dif[i]-1))/
    > ((1/v_dif[i])*(1-(1/v_dif[i]))^(v_dif[i]-1)))
  }
  chi2_95_ex = qchisq(.95, df=fail)

  if (fail==0) {
    LR_ind = 1
    chi2_95_ex = 0
  }

  if (LR_POF > chi2_95) {
    inaccurate = inaccurate + 1;
  }
  if (LR_TUFF > chi2_95) {
    inaccurate_TUFF = inaccurate_TUFF + 1;
  }
  if (LR_ind > chi2_95_ex) {
    inaccurate_ind = inaccurate_ind + 1;
  }
}
spatne = c(spatne, inaccurate/1000)
spatne2 = c(spatne2, inaccurate_TUFF/1000)
spatne3 = c(spatne3, inaccurate_ind/1000)
}

plot(sample,spatne, type="l", col="blue", xlab="sample_size",
> ylab="Probability of Type I error")
lines(sample,spatne2,col="green")
lines(sample,spatne3, col="red", type="l")
abline(h =0.05, lty=3)
legend(500,0.13, lty=c(1,1,1,3), c("POF-test", "TUFF-test",
> "Independence test", "Significance level"),
> lwd=c(2.5,2.5,2.5,2.5),col=c("blue", "green", "red", "black"))

```